

```
module decoder (IN, Y);  
  input  [1: 0] IN;           // Two binary inputs  
  output  [3: 0] Y;           // Four binary outputs  
  reg     [3: 0] Y;  
  integer  k;                 // Control (index) variable for loop  
  always @ (IN)  
    for (k = 0; k <= 3; k = k + 1)  
      if (IN == k) Y[k] = 1;  
      else Y[k] = 0;  
endmodule
```
